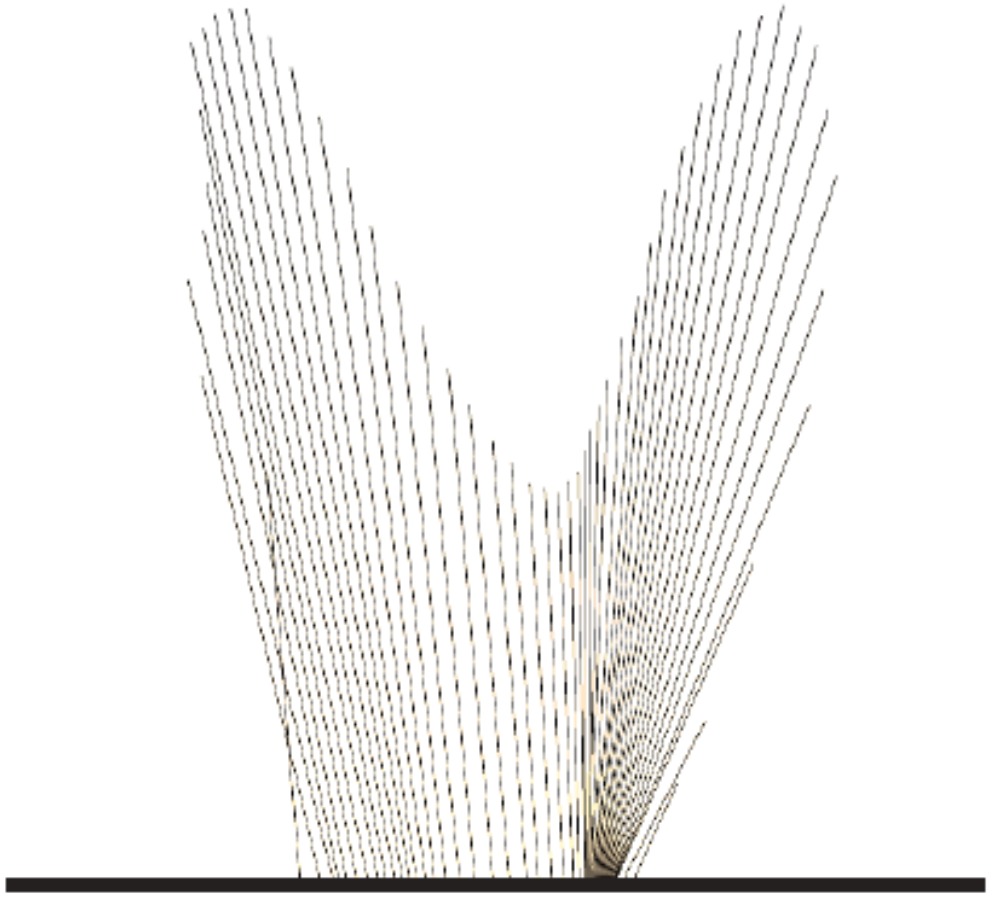
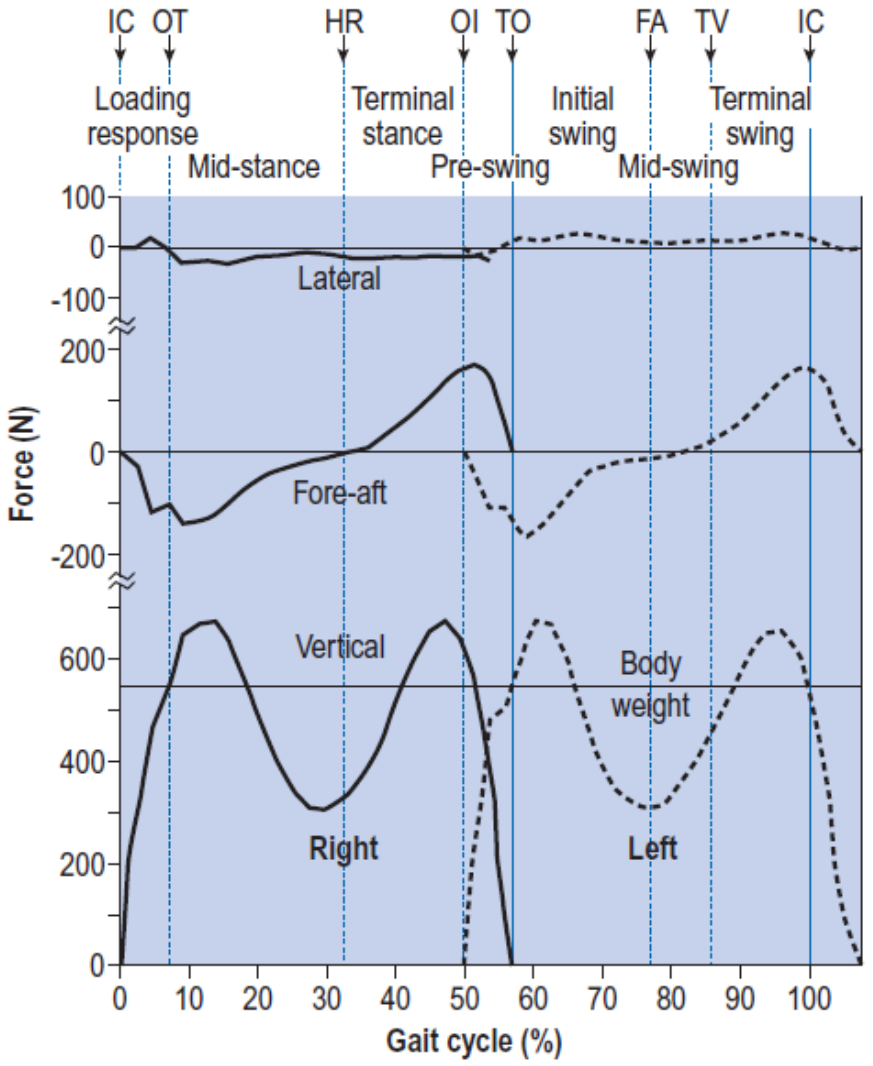
GRF





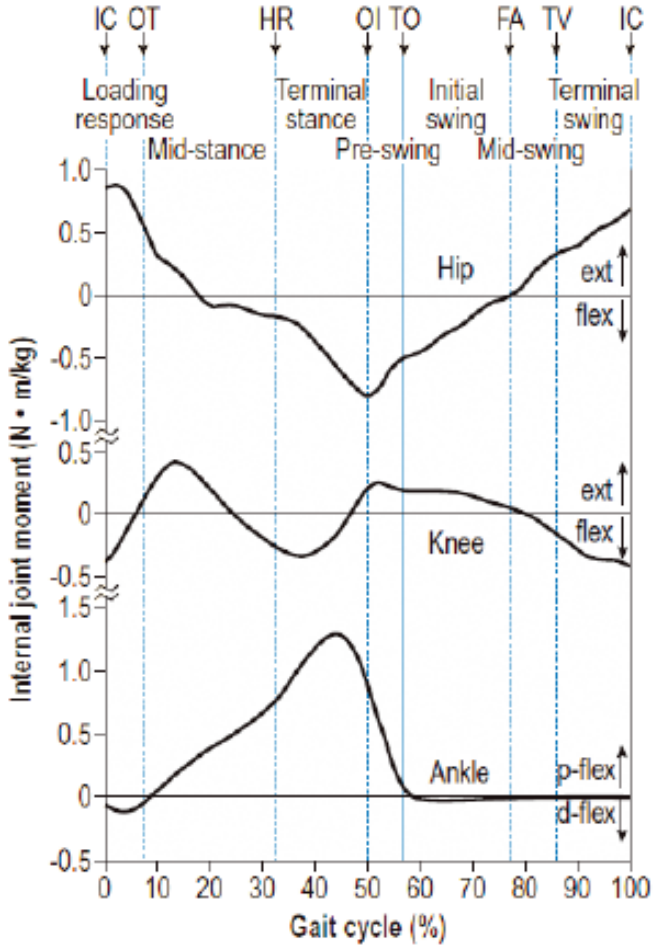


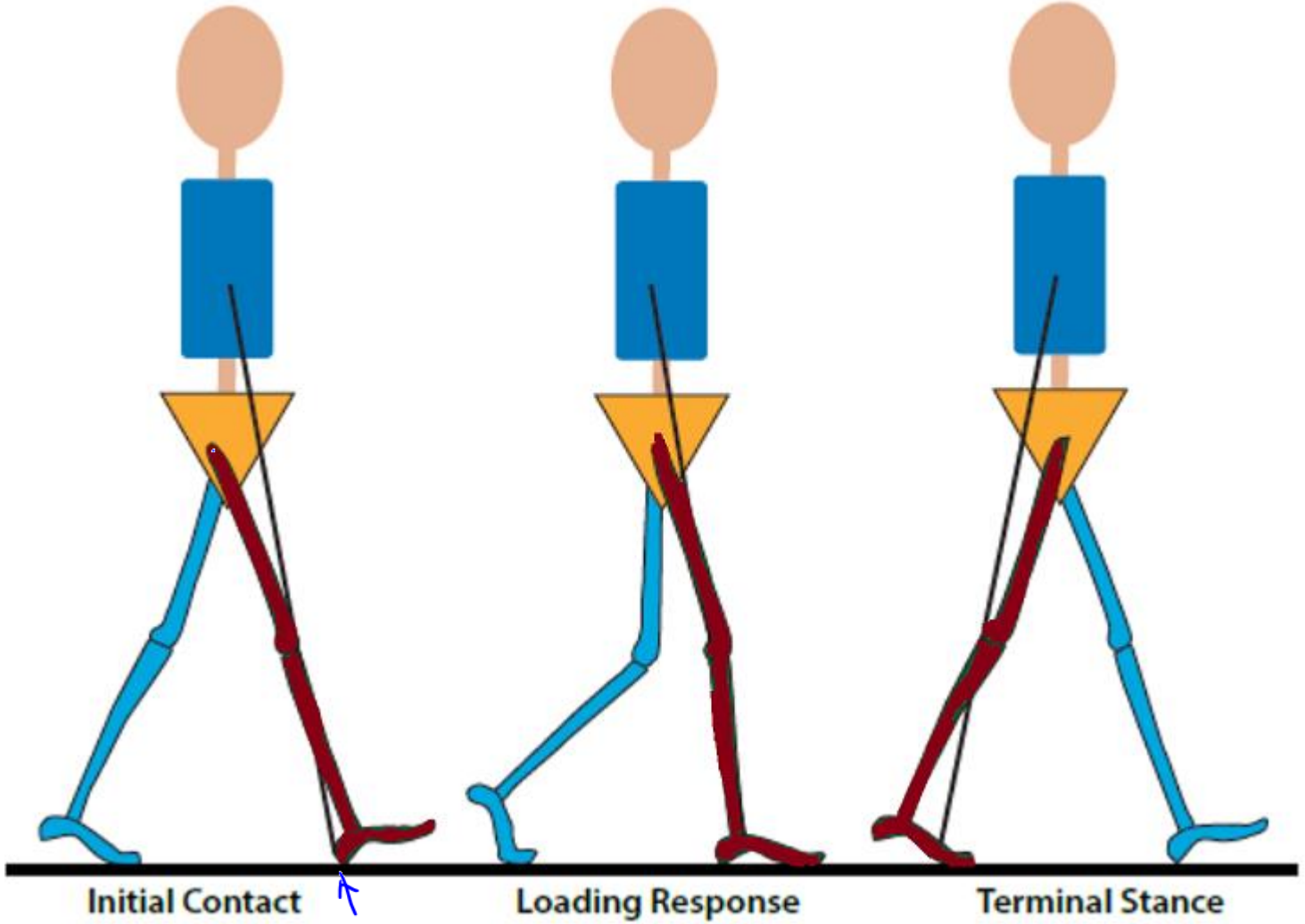
M-L

A-P

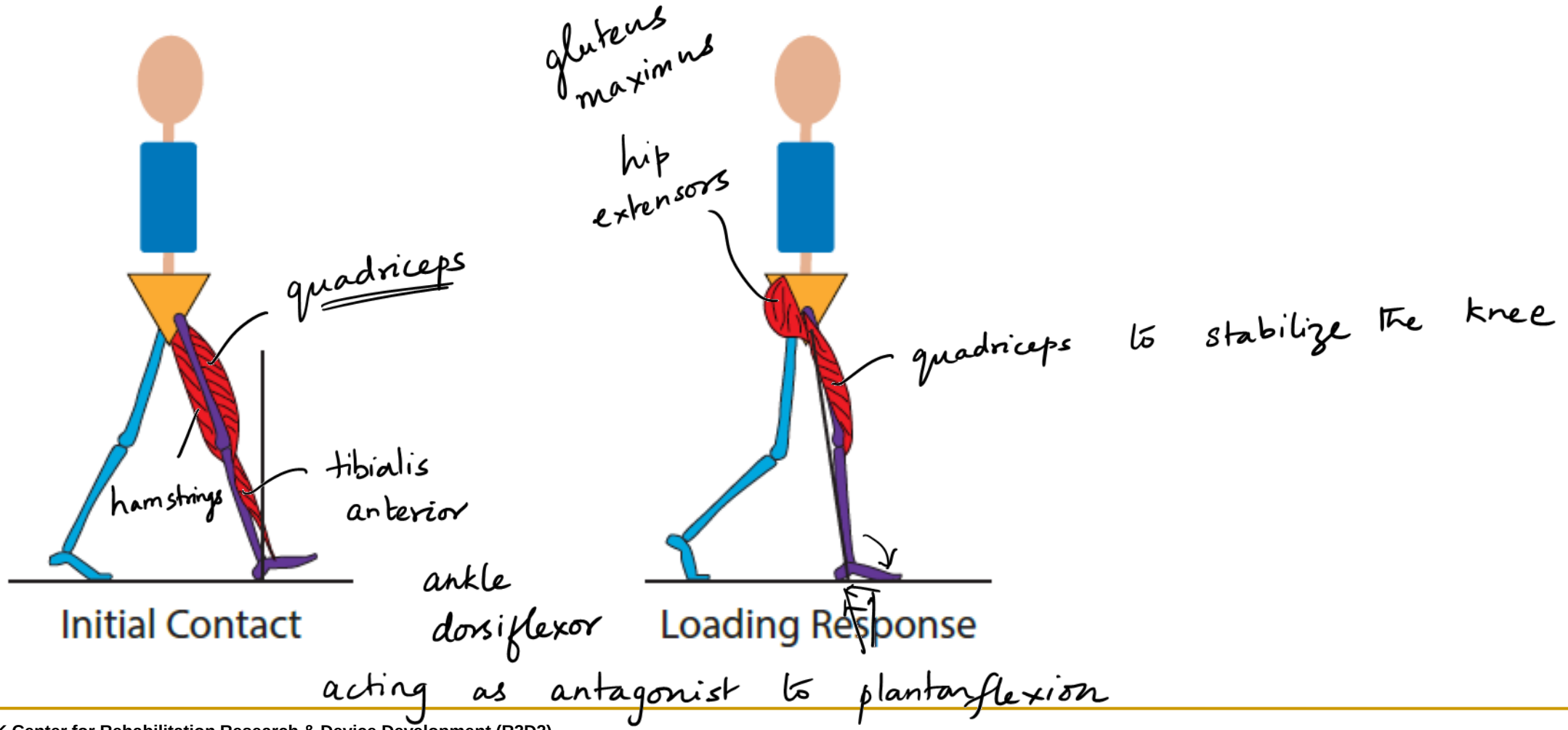
Source: Gait Analysis by Michael Whittle

Joint moments





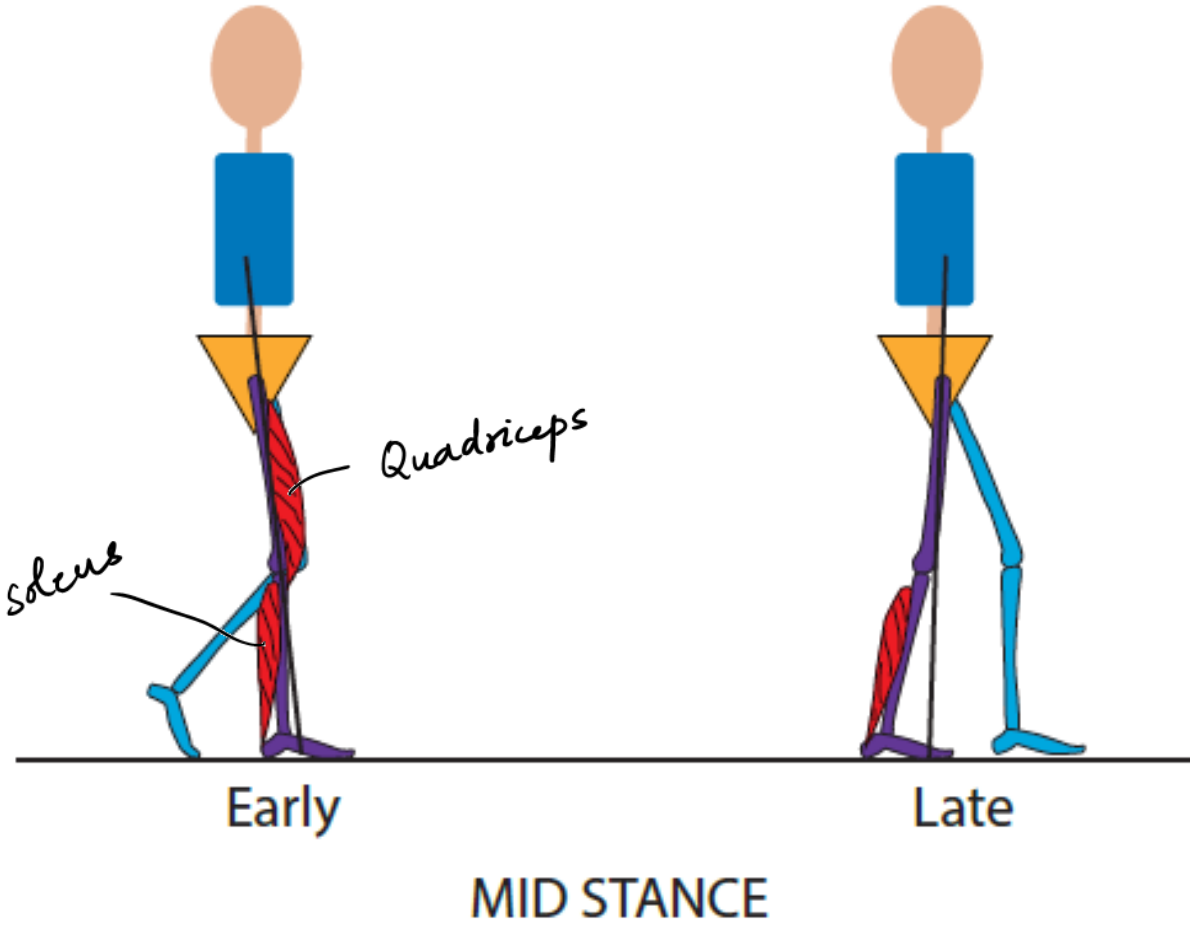
Muscle action



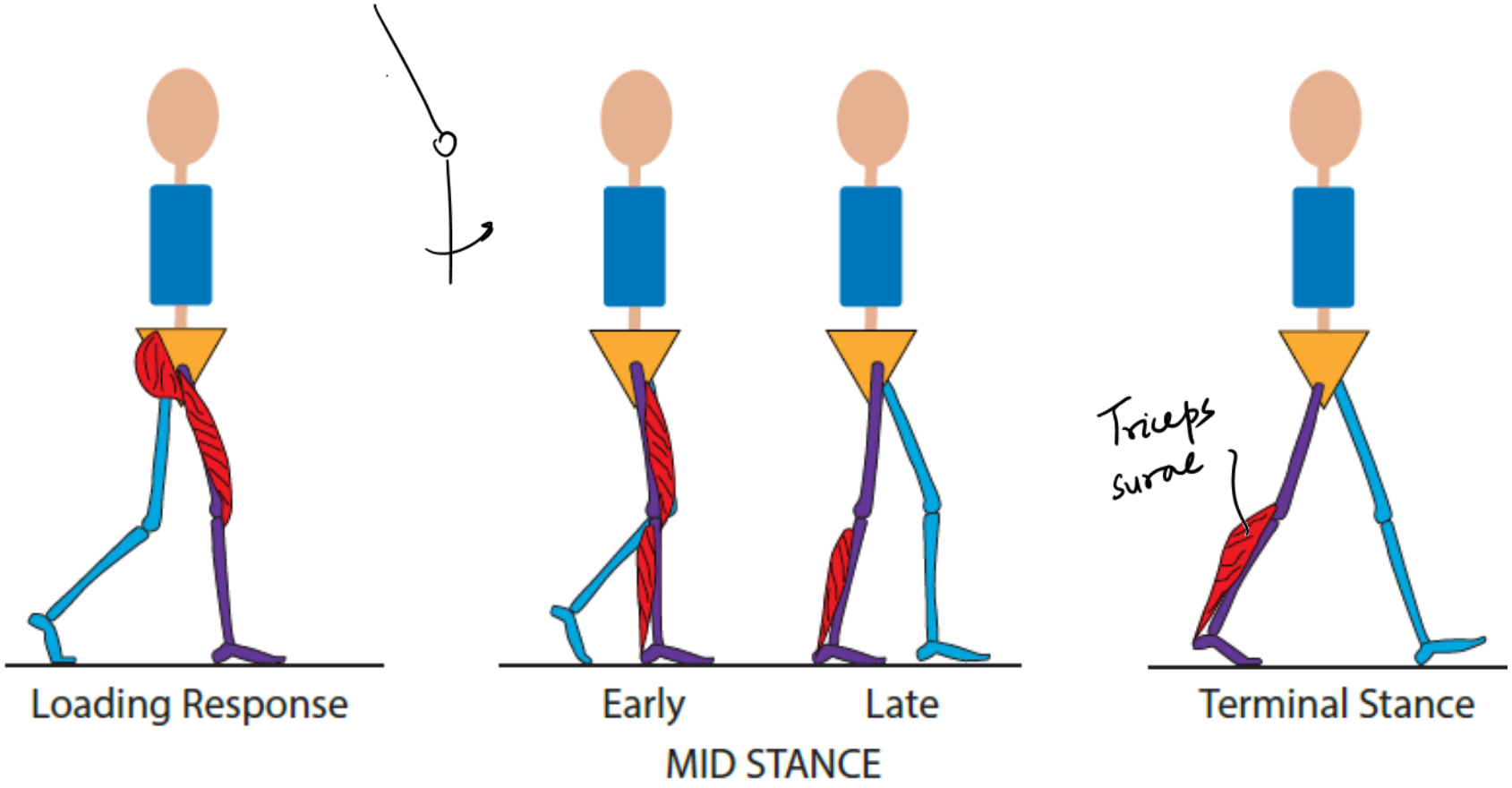
Muscle action

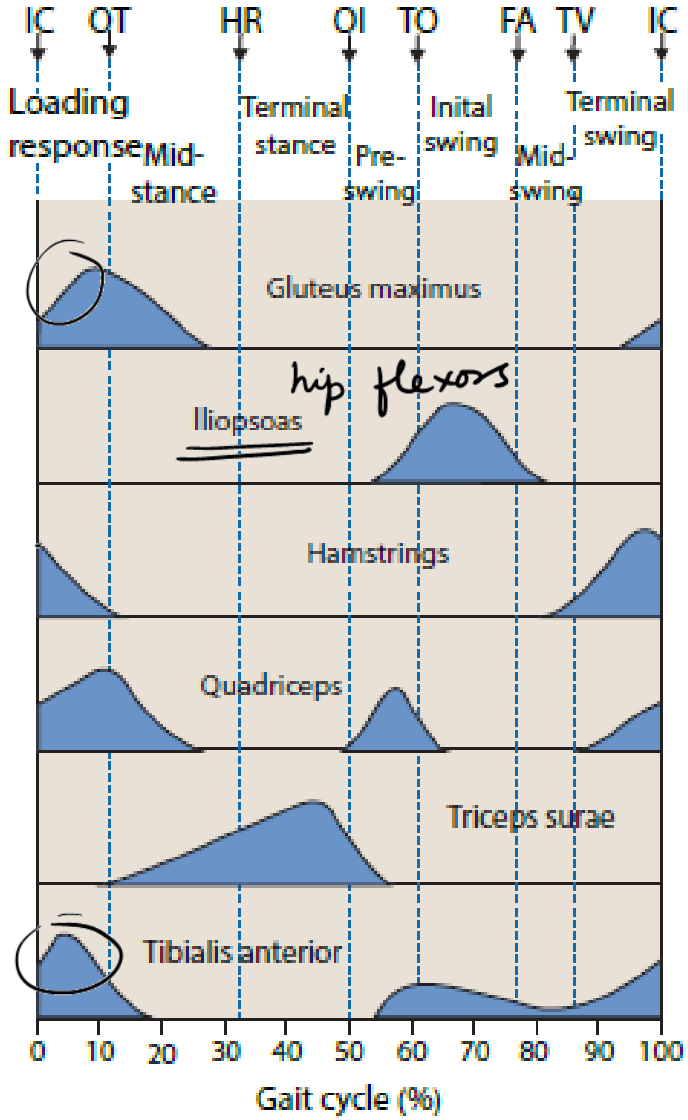
As long as the GRF falls anterior to the knee, the knee is stable

Gastroc + soleus → triceps surae



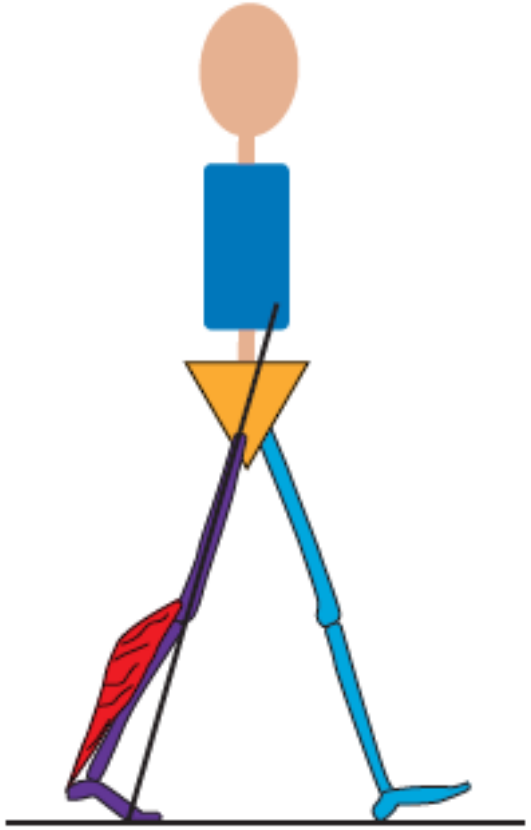
Muscle action



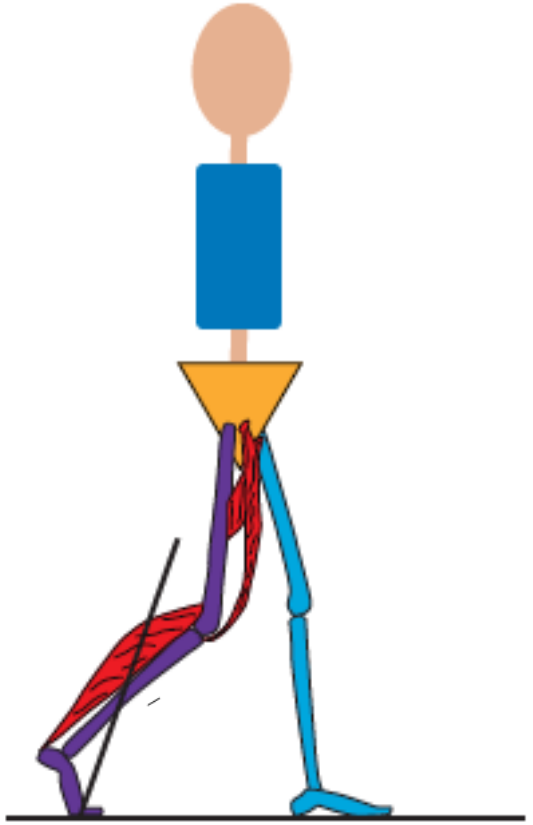


Source: Gait Analysis by Michael Whittle

Muscle action

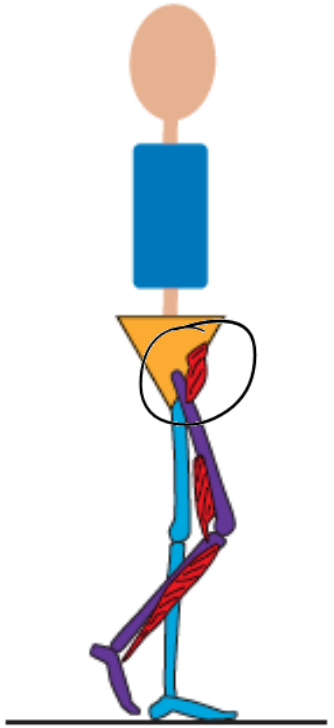


Terminal Stance

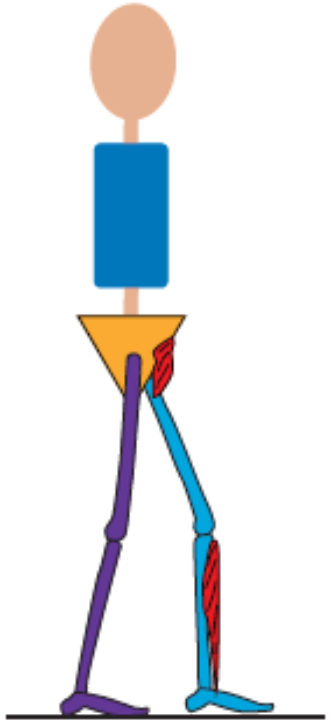


Pre-Swing

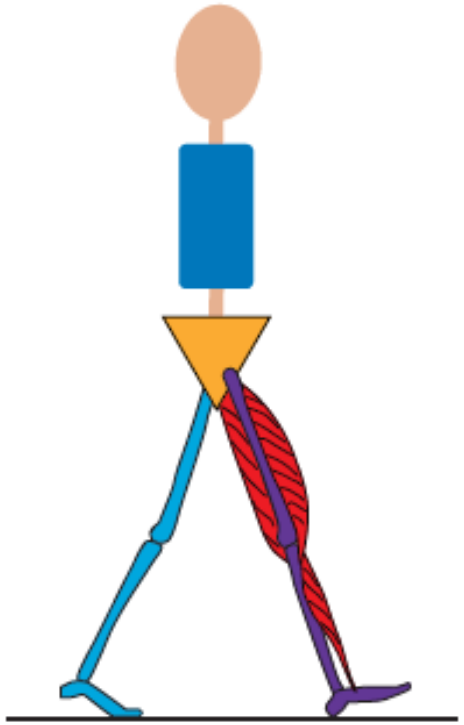
Muscle action



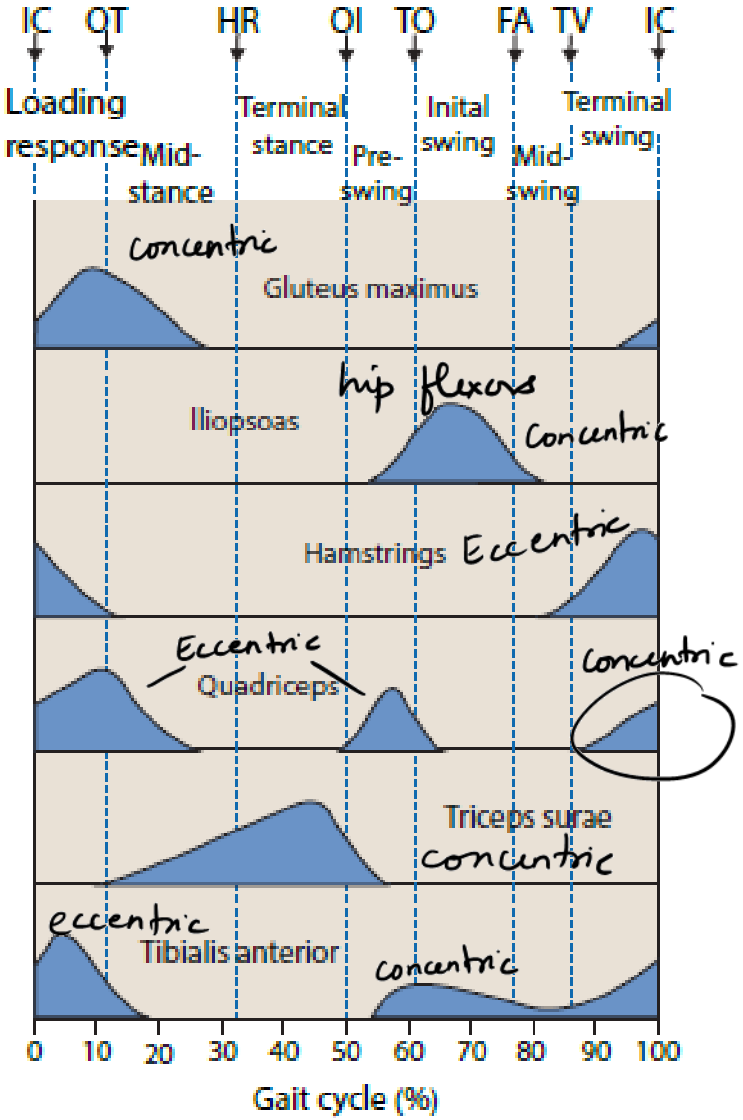
Initial Swing



Mid Swing

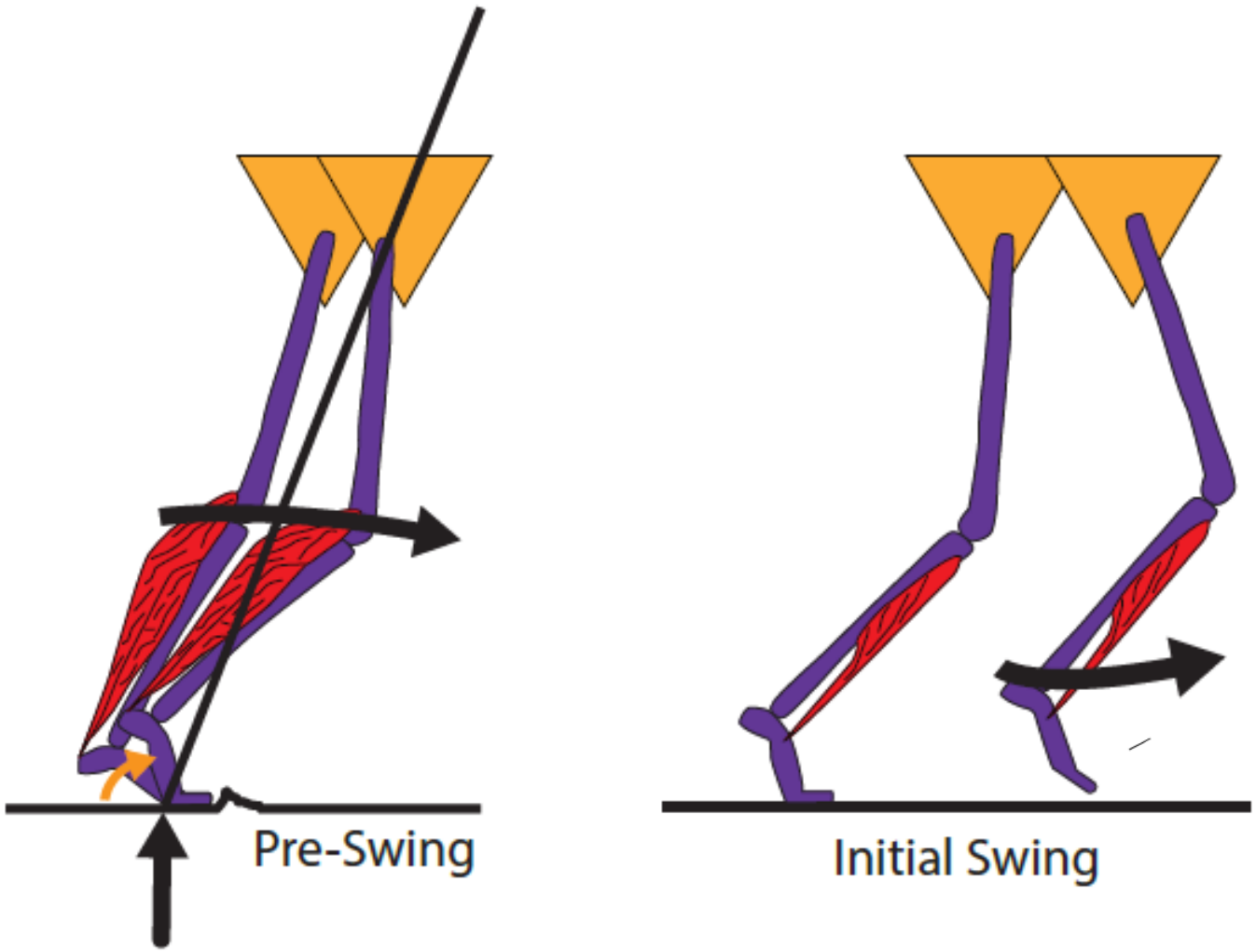


Terminal Swing

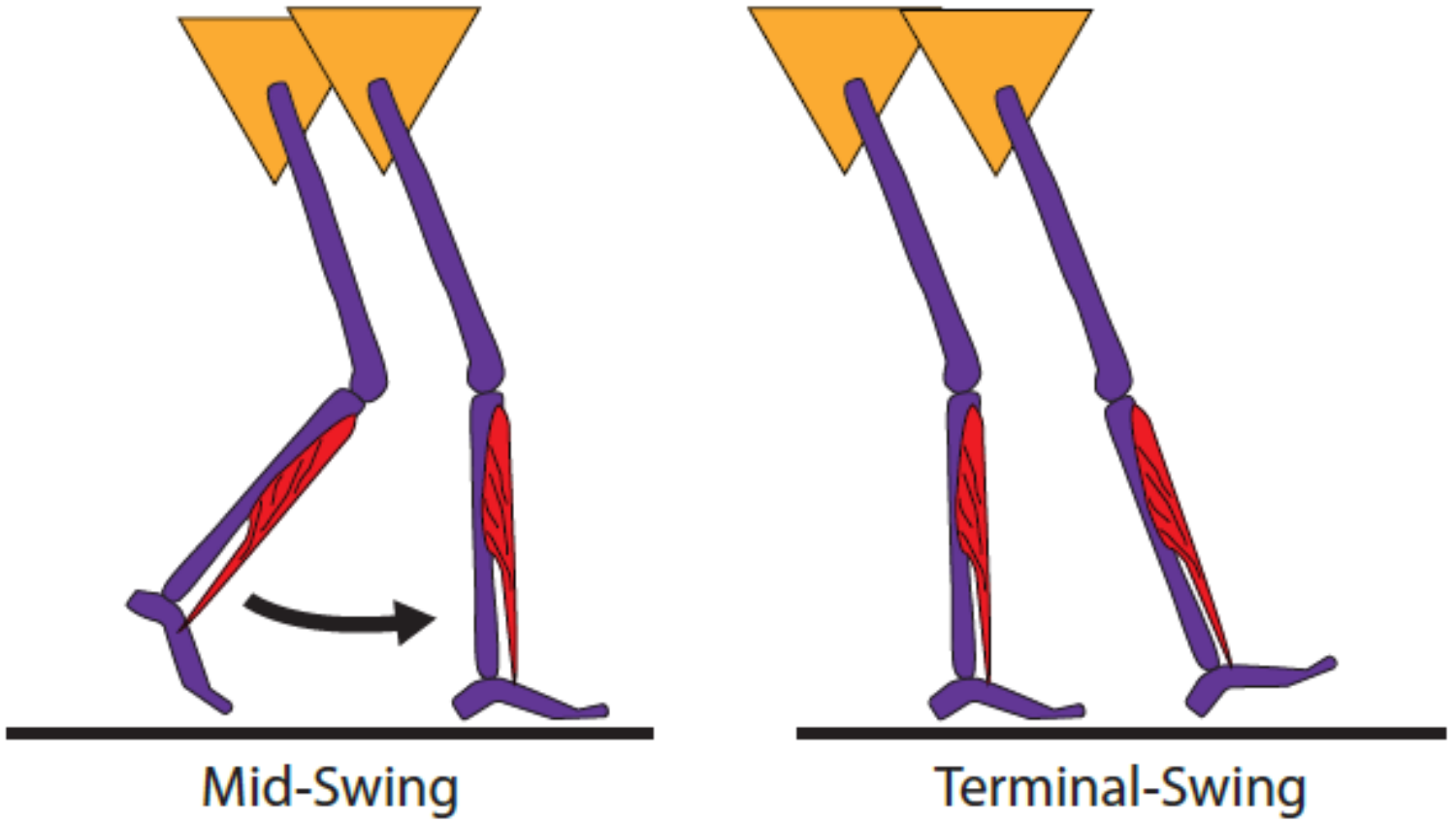


Source: Gait Analysis by Michael Whittle

Muscle action

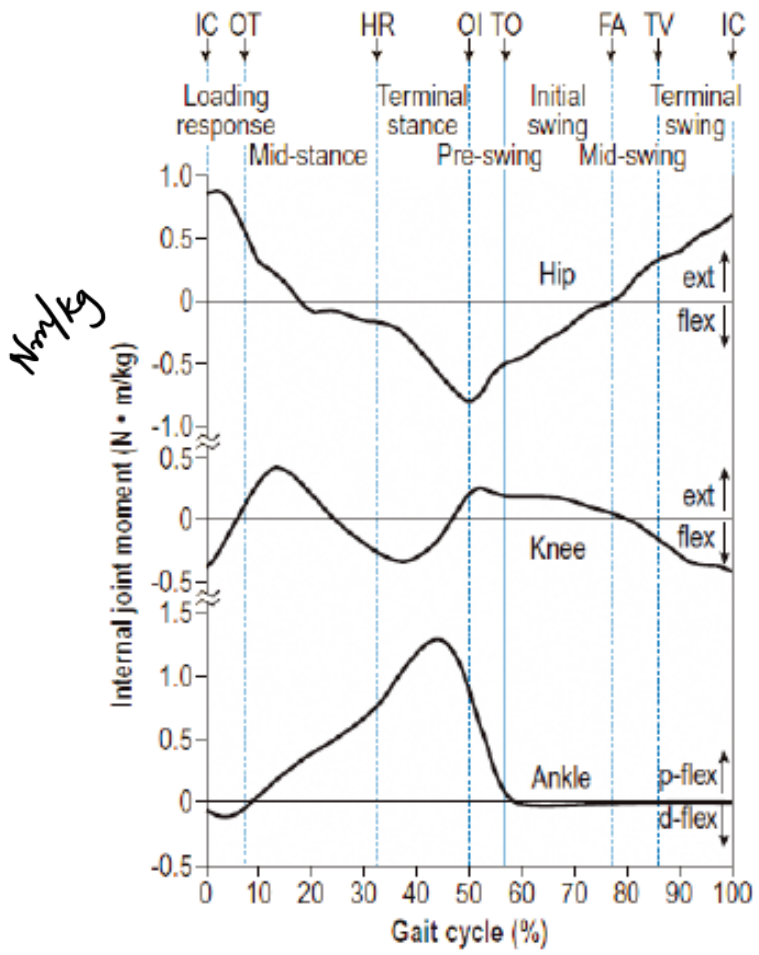


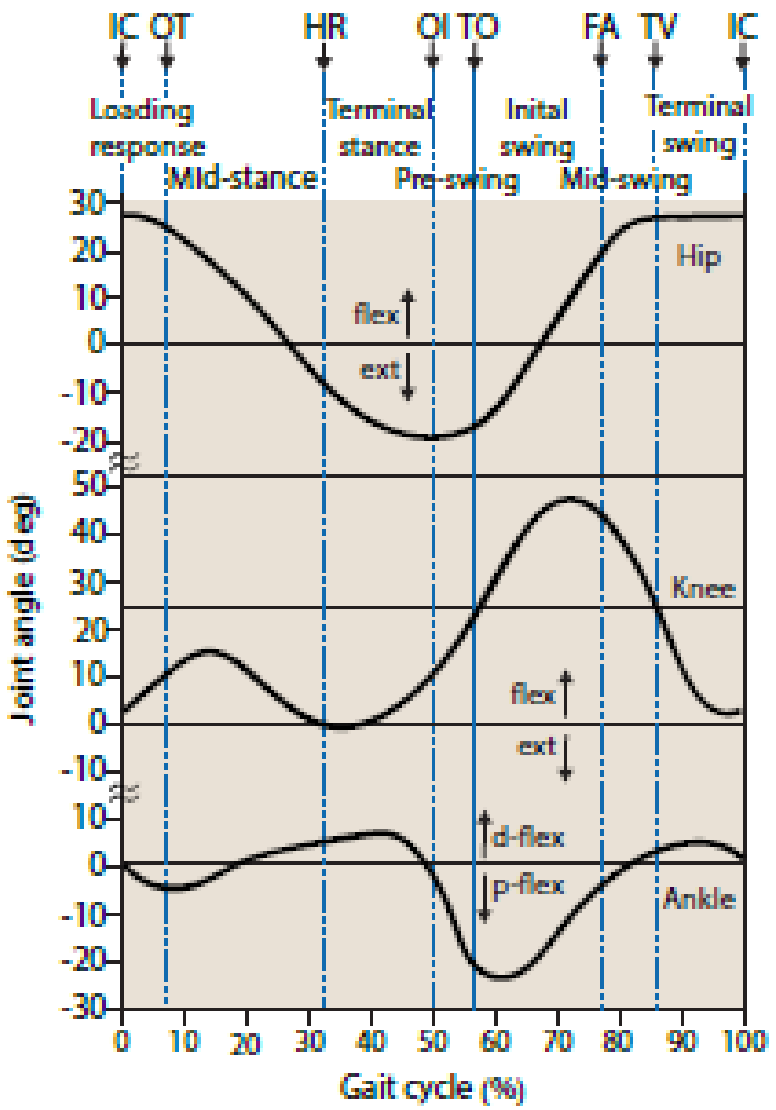
Muscle action

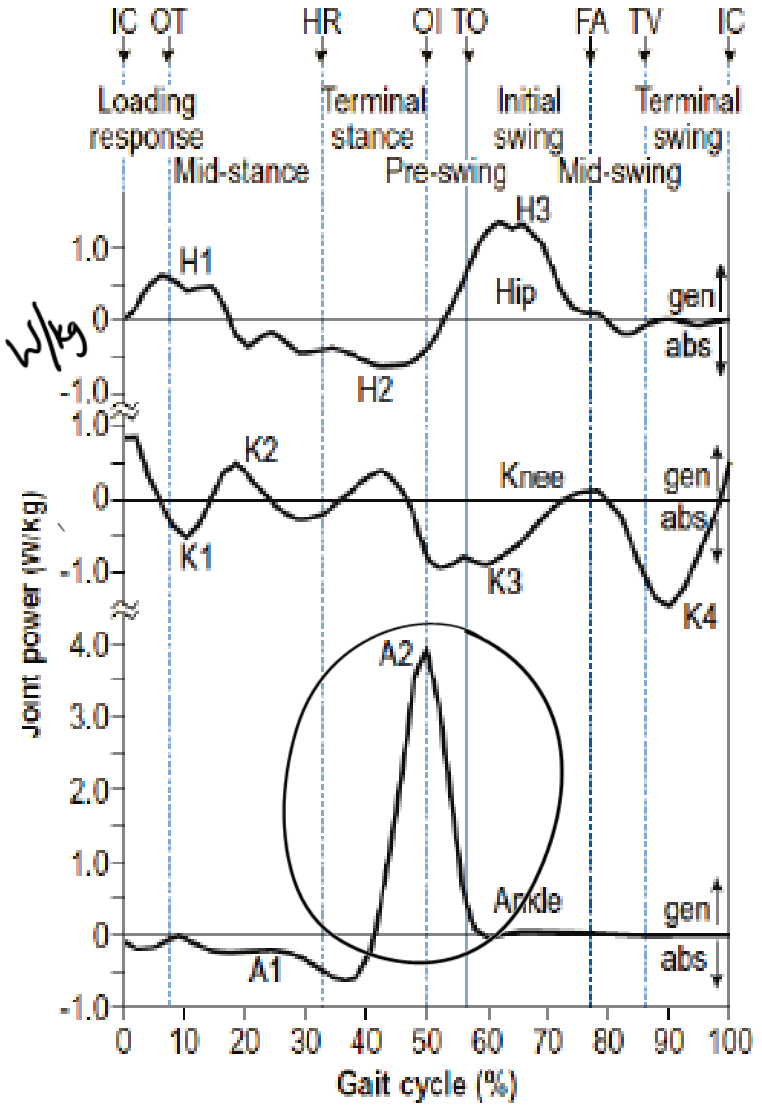


Muscle action

Joint power = $\bar{M} \cdot \bar{\omega}$



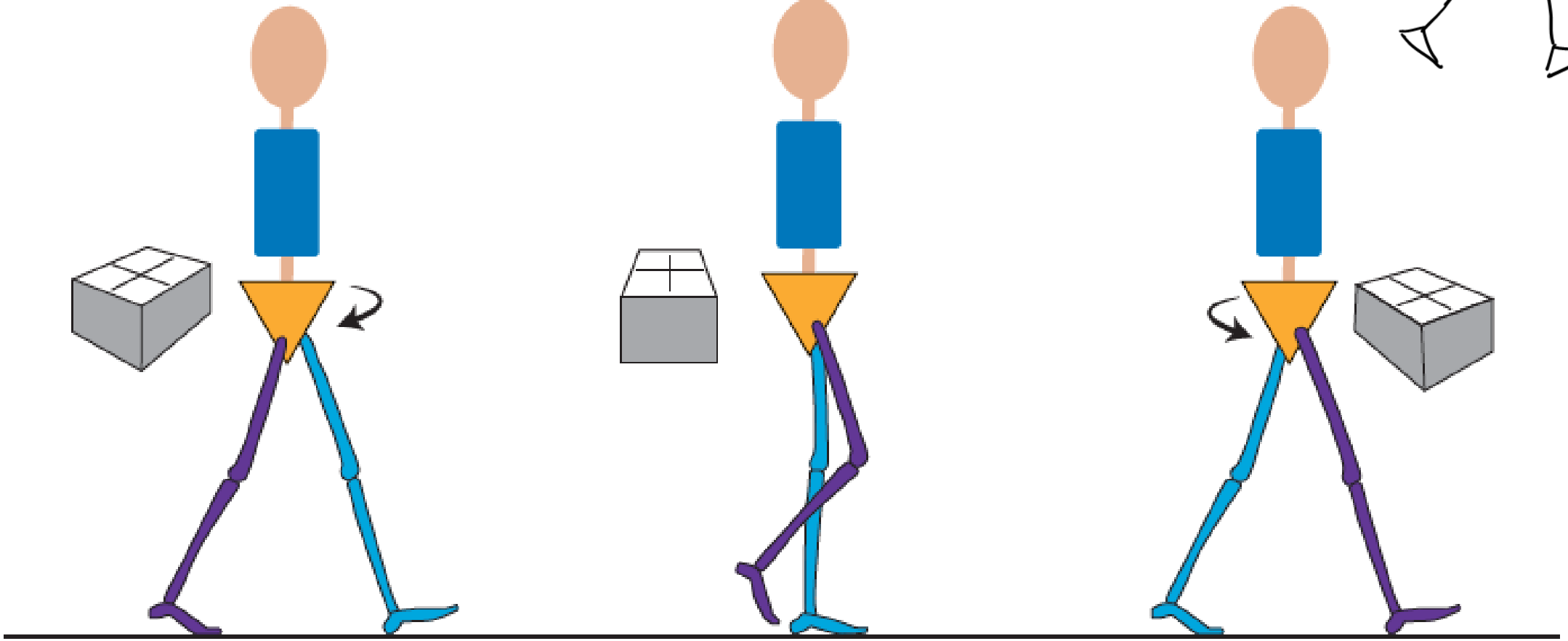


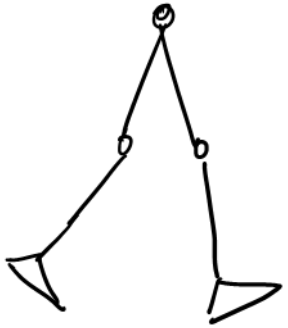


Source: Gait Analysis by Michael Whittle

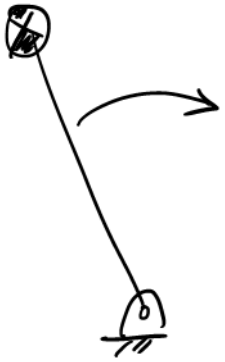
Pelvic mobility

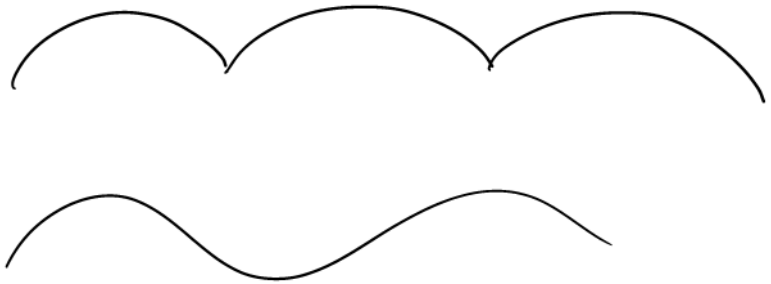
Pelvis movement in the transverse plane increases the stride length

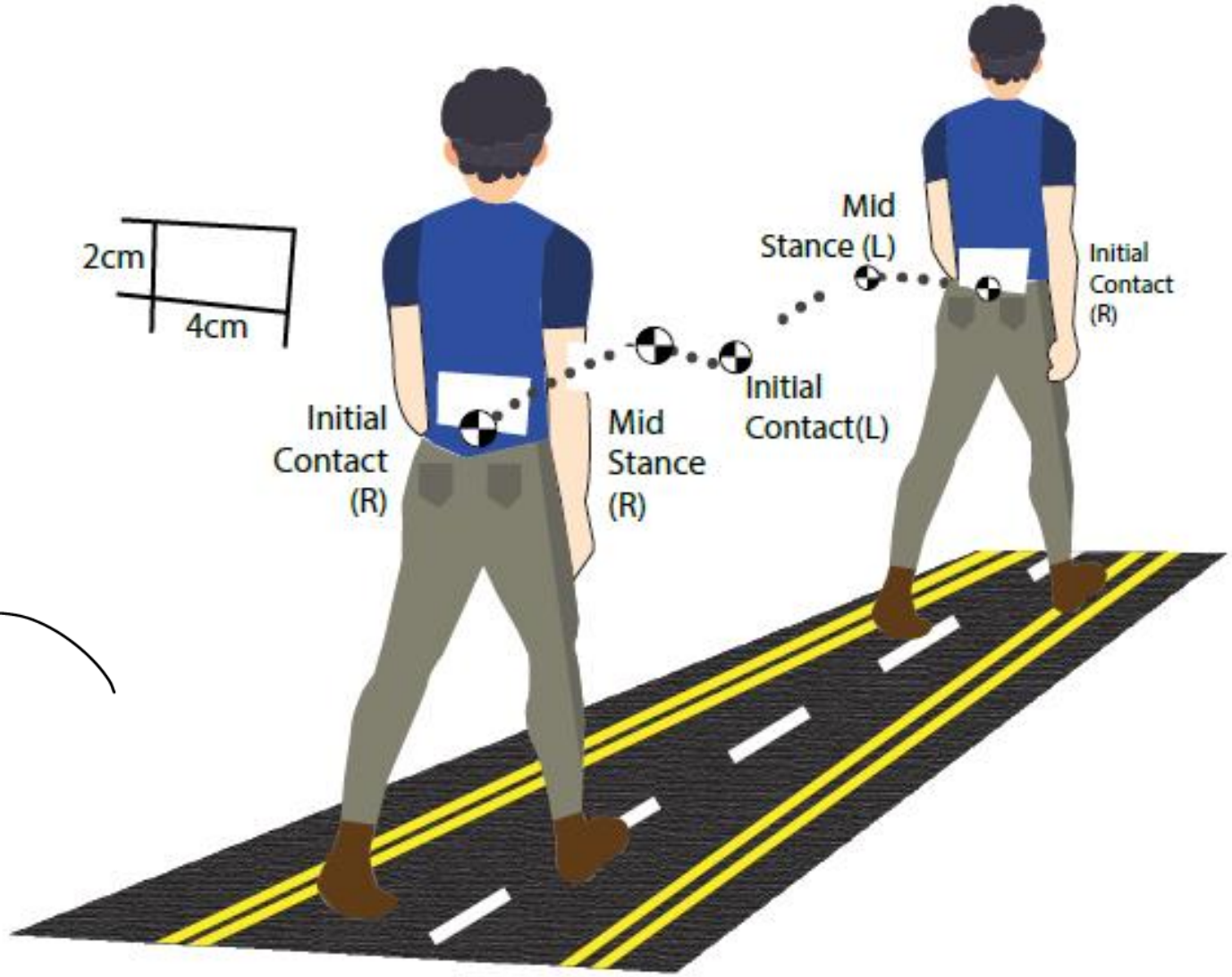




Path of CoM






2cm
4cm
Initial Contact (R)
Mid Stance (R)
Initial Contact(L)
Mid Stance (L)
Initial Contact (R)

IC : Stable weight-bearing , extended knee posture

LR : Controlled knee flexion – shock absorption

Mid-stance : Knee extension , stable weight-bearing

Stance stability is predominantly dependent on an extended knee.

Terminal stance : complete knee extension, weight bearing

Preswing : Passive knee flexion , prepare for swing

Initial swing : Knee flexion
Foot clearance for advancing the limb

Mid swing : Passive knee extension , limb advancement

Terminal swing : Knee extension (some active knee extn. may be reqd.)
Limb advancement, prepare for stance